


```

| | +-- __init__.py
| | +-- ai_automation.py      <- KnowledgeSearchRequest/Response * (NEW)
| | +-- user.py
| | +-- lead.py
| | +-- appointment.py
| | +-- clinic.py
| | +-- organization.py
| | +-- revenue.py
| | +-- audit.py
| | +-- call.py
| | +-- note.py
| | +-- task.py
| | +-- report.py
| | +-- common.py
| | +-- pagination.py
| |
| +-- services/              <- Business Logic Layer
| | +-- __init__.py
| | +-- openai_service.py    <- Gemini AI + Embedding generation * (UPDATED)
| | +-- ai_automation.py     <- RAG retriever + knowledge CRUD * (UPDATED)
| | +-- auth.py              <- JWT, token management
| | +-- lead.py
| | +-- appointment.py
| | +-- user.py
| | +-- clinic.py
| | +-- organization.py
| | +-- report.py
| | +-- revenue.py
| | +-- audit.py
| | +-- webhook.py
| |
| +-- core/                  <- Infrastructure & Cross-cutting Concerns
| | +-- __init__.py
| | +-- config.py            <- Settings (env vars)
| | +-- rate_limiter.py      <- API rate limiting (slowapi)
| | +-- cache.py             <- Redis caching
| | +-- redis_client.py      <- Redis connection
| | +-- roles.py             <- RBAC role definitions
| | +-- exceptions.py        <- Custom exception handlers
| | +-- idempotency.py       <- Idempotency key handling
| | +-- supabase_client.py   <- Supabase connection
| | +-- auth_utils.py        <- Auth helper utilities
| |
| +-- alembic/               <- Database Migrations (MOVED from root)
| | +-- env.py               <- Alembic environment setup
| | +-- script.py.mako       <- Migration template
| | +-- README
| | +-- versions/            <- Migration history
| | | +-- 88fe24ac8b12_baseline_schema.py
| | | +-- 2c6db800bb3e_add_hashed_password_to_users.py
| | | +-- 4679bc5e7f8b_add_ai_automation_tables.py
| | | +-- da46f6e2485e_add_rls_policies_to_users.py
| | +-- sql_migrations/      <- Raw SQL (MERGED from old migrations/)
| | +-- 001_initial_schema.sql
| |
| +-- scripts/               <- Utility Scripts (NEW - was scattered)
| | +-- generate_change_report_pdf.py <- PDF report generator
| | +-- generate_hash.py      <- Password hash utility
| |
| +-- tests/                 <- Test Suite
| | +-- conftest.py
| | +-- test_ai_fallback.py
| | +-- test_audit_remaining_features.py
| | +-- test_caching.py

```

```
+-- test_exception_handling.py
+-- test_idempotency.py
+-- test_jwt_revocation.py
+-- test_rate_limiter.py
+-- test_refresh_token.py
+-- test_soft_delete_and_pagination.py
+-- test_webhooks.py
```

deploy/ — Deployment Configs

```
deploy/
+-- nginx/
   +-- nginx.conf          <- Nginx reverse proxy config
```

docs/ — All Documentation (RENAMED from documents/)

```
docs/
+-- BACKEND_README.md      <- Backend dev guide
+-- QUICKSTART.md          <- Quick setup guide
+-- DOCUMENTATION_URDU.md  <- Urdu documentation
+-- CHANGE_REPORT.pdf      <- Change history
+-- AI_Automation_Implementation_Report.pdf
+-- AI_Automation_Scope_Dental_CRM.pdf
+-- AI_Features_Testing_Guide.pdf
+-- Dental_CRM_AI_Automation_Interactive_Platform_Specification.md
+-- Dental_CRM_Backend_Audit_Report.pdf
+-- Dental_CRM_Backend_Technical_Learning_Guide.md
+-- Dental_CRM_Complete_File_Guide.pdf
+-- Access_defined_Dental_CRM.docx
+-- folderstructure-EXPLANATION.docx
```

■ Changes Summary — Old vs New

Old Location	New Location	Reason
alembic/ (root)	backend/alembic/	App ke saath hona chahiye
alembic.ini (root)	backend/alembic.ini	alembic/ ke paas
backend/migrations/	backend/alembic/sql_migrations/	Single migrations folder
requirements.txt (root)	■ Deleted	Duplicate tha
backend/requirements.txt	backend/requirements.txt ■	Numpy + pgvector add kiya
generate_hash.py (root)	backend/scripts/	Utility scripts ek jagah
backend/generate_change_report_pdf.py	backend/scripts/	Utility scripts ek jagah
deploy/Dockerfile	Dockerfile (root)	Docker standard practice
documents/	docs/	Proper naming convention
backend/*.md, *.pdf	docs/	Docs ek jagah
backend.zip (root)	■ Deleted	Compiled artifact — git mein nahi hona chahiye

■ Commands — Developer Reference

Backend run karo:

```
cd backend
uvicorn app.main:app --reload --port 8000
```

Migrations run karo (backend/ folder se):

```
cd backend  
alembic upgrade head
```

Tests run karo:

```
cd backend  
pytest tests/
```

Requirements install karo:

```
cd backend  
pip install -r requirements.txt
```

■ Docker (Root se chalao)

Root folder se (jahan Dockerfile hai):

```
docker-compose up --build
```